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NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)
Sixth Semester B.Tech (CBCS) Degree Examinations

May 2025

RI2214-1- VIRTUAL INSTRUMENTATION

(For RI)

Duration: 3 Hours

Max. Marks: 100

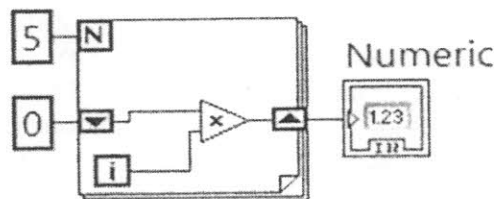
Note:

Part – A: Multiple Choice Questions: Answer all **Twenty** questions in the **OMR Sheet** provided. Each question carries equal marks.

Part – B: Descriptive Answer Questions: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II** each and **One full** question from **Unit – III**.

PART - A: MULTIPLE CHOICE QUESTIONS**20 Marks**

- Which of the following is NOT a typical advantage of using Virtual Instruments (VIs) over traditional hardware instruments?
A) Flexibility in design and functionality
B) Improved precision of measurement
C) Ability to integrate multiple hardware interfaces
D) Lower cost of development and operation
- Which of the following is true about file operations in LabVIEW?
A) LabVIEW file operations cannot handle binary data
B) File I/O operations require external software to manage data storage.
C) LabVIEW allows reading and writing text and binary files.
D) File operations in LabVIEW only work with data from virtual instruments.
- You are building a VI that records temperature over time and displays the data on a graph. Which of the following would you use to show continuous temperature readings over time in LabVIEW?
A) Graph
B) Chart
C) Numeric Indicator
D) String Indicator
- What is the value of Numeric Indicator



- A) 0
B) 1
C) 5
D) 6
- In LabVIEW, when creating a sub-VI, which of the following steps is NOT required?
A) Define the front panel controls and indicators
B) Create a connector pane to define input and output terminals
C) Save the sub-VI to a new file
D) Use a loop to repeat operations in the sub-VI
- Which of the following is true about Express VIs in LabVIEW?
A) Express VIs requires extensive custom programming to function properly
B) Express VIs are prebuilt VIs designed to simplify complex tasks such as data acquisition and analysis.
C) Express VIs can only be used in the front panel of a VI.
D) Express VIs cannot be customized by the user.

7. You have created a complex temperature control algorithm inside LabVIEW VI. Now, you want to reuse this algorithm in multiple projects without rewriting the code. What is the best way to achieve this in LabVIEW?
- A) Copy and paste the block diagram into every new VI B) Save the code as a Sub-VI and use it in other VIs
C) Use local variables to pass data between VIs D) Convert the VI into an Express VI
8. You are building a control system where a motor should start only if the temperature exceeds 50°C. Otherwise, the motor should remain off. Which LabVIEW structure is best suited for implementing this logic?
- A) For Loop B) While Loop
C) Case Structure D) Shift Register
9. Which of the following statements is TRUE about auto-indexing in LabVIEW?
- A) It is used to control loop execution speed B) It automatically iterates through array elements inside loops
C) It stores array data in a file D) It prevents loops from executing
10. In LabVIEW, what is the function of the "Replace Array Subset" function?
- A) Inserts a new value at a specified index, shifting other elements B) Deletes an element from an array
C) Replaces an existing value at a specified index without shifting elements D) Merges two arrays into one
11. You need to create a 100×100 2D array and initialize all elements to zero. Which LabVIEW function is best suited for this task?
- A) Build Array B) Array Subset
C) Initialize Array D) Insert Into Array
12. You are plotting real-time data from a temperature sensor and need to display values continuously over time without losing previous data. Which LabVIEW graph type should you use?
- A) XY Graph B) Waveform Graph
C) Waveform Chart D) 3D Graph
13. Which type of measurement requires a differential input configuration?
- A) Measuring a voltage from a single-ended sensor B) Measuring the difference between two signal voltages
C) Measuring digital pulses D) Measuring a thermocouple temperature
14. What is the main purpose of the DAQ Assistant in LabVIEW?
- A) To manually program hardware-level DAQ settings B) To simplify the configuration of NI DAQ hardware without coding
C) To convert analog data into digital format D) To generate random signals for testing
15. What is the main difference between continuous and finite data acquisition?
- A) Continuous acquisition collects data indefinitely, while finite acquisition stops after a fixed number of samples B) Finite acquisition is used for real-time monitoring, while continuous acquisition is not
C) Continuous acquisition requires hardware triggering, while finite acquisition does not D) Finite acquisition is used for high-speed applications, while continuous acquisition is for slow-speed applications
16. What is the advantage of generating LabVIEW code from a MAX DAQmx Task?
- A) It eliminates the need to manually configure DAQ settings in LabVIEW B) It speeds up execution time by optimizing the compiled code
C) It allows LabVIEW to control multiple DAQ devices without hardware D) It prevents aliasing in sampled signals
17. The maximum voltage range of NI myDAQ's analog input is
- A) ±5V B) ±10V
C) 0 to 5V D) 0 to 10V
18. The sampling rate of NI myDAQ depends on
- A) Signal frequency B) Number of channels used
C) Both A and B D) None of the above

19. Which of the following methods allows Python to communicate with LabVIEW?
- A) LabVIEW Python Node B) TCP/IP
C) Shared Variables D) All the above
20. Which of the following is NOT a valid reason to use Python with LabVIEW?
- A) Python can perform advanced data processing and machine learning B) Python can replace LabVIEW's real-time execution engine
C) Python allows integration with libraries not available in LabVIEW D) Python enables scripting automation for repetitive tasks

PART - B: DESCRIPTIVE ANSWER QUESTIONS

Unit – I		Marks	BT*	CO*	PO*
1.	a) Differentiate between Virtual Instruments and Traditional Instruments in terms of functionality, flexibility, and cost-effectiveness	04	L*2	1	2,3,5
	b) What are the essential components of a LabVIEW environment? Describe the function of the Front Panel and Block Diagram	04	L2	1	2,3,5
	c) Describe the use of graphs and charts in LabVIEW. How do they enhance data visualization?	08	L2	2	2,3,5
2.	a) Explain the importance of file operations in LabVIEW. How are data logs managed	04	L2	2	2,3,5
	b) What is the difference between a Sub-VI and an Express VI? Provide an example use case for each	04	L2	2	2,3,5
	c) Develop a VI that takes an integer as input and determines whether it is even or odd. Display the result as a Boolean indicator	08	L3	1	2,3,5
3.	a) You need to design a VI that calculates the cumulative sum of numbers entered by the user. How would you use shift registers to achieve this?	04	L3	2	2,3,5
	b) Simulate an LED that blinks at a user-defined interval. Use a while loop with a delay and a Boolean indicator to represent the LED	04	L3	1	2,3,5
	c) What is the role of Shift Registers in LabVIEW? How do they differ from Feedback Nodes?	04	L2	2	2,3,5
	d) Describe the use of event structures with an example program in LabVIEW.	04	L3	1	2,3,5
Unit – II					
4.	a) Explain the difference between one-dimensional, two-dimensional, and multi-dimensional arrays in LabVIEW.	04	L2	3	2,3,5
	b) What are the advantages of using array functions in LabVIEW over traditional loops?	04	L2	3	2,3,5
	c) Explain the concept of finding a common ground and measuring signal differences in a DAQ system.	08	L2	4	2,3,5
5.	a) How does NI-DAQmx differ from traditional DAQ programming? Discuss performance, flexibility, and ease of use.	04	L2	4	2,3,5
	b) Develop a LabVIEW application that acquires and processes signals from multiple sensors and logs the data into a file.	08	L3	4	2,3,5
	c) How can you insert, delete, and replace elements in a LabVIEW array?	04	L2	3	2,3,5
6.	a) Explain how auto-indexing can be controlled using shift registers. Provide a practical example.	04	L2	3	2,3,5
	b) Discuss timed structures in LabVIEW. How can they be used for real-time applications?	04	L2	4	2,3,5

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| c) Compare Case Structures and Event Structures in LabVIEW. | 04 | L3 | 3 | 2,3,5 |
| d) Describe how waveform data types handle time information. Why are they useful in signal processing? | 04 | L3 | 4 | 2,3,5 |

Unit – III

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| 7. a) Define data acquisition and list the types of signals that can be acquired using NI myDAQ. | 04 | L2 | 5 | 2,3,5 |
| b) What are the benefits of integrating Python with LabVIEW? | 04 | L2 | 5 | 2,3,5 |
| c) Explain how to generate a PWM signal using NI myDAQ and control an external device. | 08 | L3 | 5 | 2,3,5 |
| 8. a) What is the difference between a child class and a parent class in LabVIEW OOP? | 04 | L2 | 5 | 2,3,5 |
| b) How do you acquire an image from a USB camera in LabVIEW? | 04 | L2 | 5 | 2,3,5 |
| c) What are LINX and LIFA in LabVIEW? Compare their advantages and limitations. | 08 | L2 | 5 | 2,3,5 |

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
